

Yichao Jin, Ph.D.

School of Economic, Political and Policy Sciences
The University of Texas at Dallas, Richardson, TX 75080

<https://yichao2022.github.io> | [ORCID](#) | Yichao.Jin@UTDallas.edu

Health Decision Science / Discrete Choice Experiments / Medical Decision Making / Patient Preferences

Education

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| University of Texas at Dallas | May 2026 |
| Ph.D. in Public Policy and Political Economy | |
| <ul style="list-style-type: none">- Dissertation: “Intertemporal Choice in Vaccination Timing: Evidence from a Discrete Choice Experiment in Wuhan” — ProQuest- Committee: Dohyeong Kim (Chair), Richard Scotch, Soyoung Kwon, Khai Chiong.- Dean of Graduate Education Dissertation Research Award (2025). | |
| University of Texas at Dallas | May 2026 |
| M.S. in Social Data Analytics and Research | |
| University of Queensland | 2019 |
| Master of Development Economics | |
| University of California, Riverside | 2017 |
| B.A. in Economics | |

Research Agenda

My research develops structural discrete choice models to understand how individuals make health decisions under institutional and behavioral constraints. I combine Discrete Choice Experiments with structural econometrics (mixed logit, latent class models) to measure how waiting time, institutional trust, and perceived costs shape vaccination behavior and medical decision-making. I also develop Behavioral Digital Twins—causally-constrained LLM-based agents calibrated to empirical choice frontiers—for pre-screening health interventions before field deployment.

Working Papers

- “Empirical-Frontier Regularization for LLM Synthetic Agents: A Preference-Anchored Framework”**
Working Paper (Target: J. Biomedical Informatics) — [SSRN Link](#). DCE-grounded empirical-frontier regularization framework for LLM synthetic agents. Using a convex anchoring rule, reduces waiting-time monotonicity violations from 62.5% to 4.2% and improves Spearman rank correlation from -0.024 to 0.952 . Presented at AIE Summer Conference 2026, DAISY 2026 (ACM CHIL 2026), and AIXPH 2026.
- “Waiting Time as a Behavioral Barrier to Vaccination Uptake: Nonlinear Heterogeneity by Institutional Trust”**
Under Review at Economic Analysis and Policy — [Link to Full Draft](#)
 - Structural DCE framework measuring how temporal barriers and institutional trust interact in vaccination decisions (N=1,027).

- Identifies nonlinear heterogeneity in waiting time sensitivity, with direct implications for vaccine campaign design.
3. **“Value Contamination in Policy Simulation: Measuring How LLM Policy-Value Orientations Shape Predicted Administrative Burden Effects”**
Working Paper — [SSRN Link](#). Develops an audit framework for value sensitivity in LLM-generated policy simulations; evaluates nine LLMs on an administrative burden in vaccination testbed.
 4. **“The Price of Waiting: Evidence on Cash–Time Trade-Offs in Vaccination Time Discount”**
Manuscript in Preparation (Expected submission: August 2026) — Targeting *Social Science & Medicine*.

Research Experience & Affiliations

Doctoral Researcher, Spatial Health AI Research Partnership (SHARP), UT Dallas

- Leading interdisciplinary research on the intersection of artificial intelligence, geospatial analytics, and health policy.
- Managing large-scale datasets and developing computational simulations (R, Python) to evaluate pandemic preparedness.
- Leading development of the Behavioral Digital Twin (BDT) framework, integrating DCE data with causally-constrained LLM agents for synthetic policy simulation.

Graduate Researcher, Behavioral Health Economics Laboratory, UT Dallas

- Conducted high-salience behavioral experiments (N=1,027) analyzing risk-weighted health decisions and institutional trust.

Conference Presentations

- **AIxPH 2026 (1st Annual Conference on AI and Public Health, Baltimore, MD)**: Empirical-Frontier Regularization for LLM Synthetic Agents — Poster Presentation.
- **DAISY 2026 Workshop at ACM Conference on Health, Inference, and Learning (CHIL) 2026**: Behavioral Digital Twins — A Causally-Faithful Generative Framework.
- **AI and Economics (AIE) Summer Conference 2026**: Empirical-Frontier Regularization for LLM Synthetic Agents.
- **APPAM Fall Research Conference 2025** (Seattle, WA): Estimating Time Discount Rate for Vaccination.
- **UT Dallas Graduate Research Symposium 2025**: Modeling Vaccination Decisions with Hyperbolic Discounting.

Technical Skills & Certifications

Econometrics: Mixed Logit (MXL), Latent Class Models (LCM), SEIR Modeling, Causal Inference.

AI & ML: PyTorch, Transformers (HuggingFace), LLM APIs (OpenAI, Anthropic), LangChain, Causal Machine Learning.

Software: Stata (Advanced), R, Python, MATLAB, LaTeX, SQL, GIS.

Certifications: Stanford Artificial Intelligence Graduate Certificate; CITI Human Subjects Protection.

Honors & Awards

- **2025** Dean of Graduate Education Dissertation Research Award, UT Dallas
- **2025** Betty & Gifford Johnson Graduate Travel Award, UT Dallas
- **2016** Omicron Delta Epsilon, International Honor Society for Economics

REFERENCES

Dohyeong Kim (Chair)

University of Texas at Dallas
Senior Associate Dean of Graduate Education
Robert E. Holmes Jr. Professor of Public Policy,
GIS & Social Data Analytics
Team Lead, SHARP
dohyeong.kim@utdallas.edu

Soyoung Kwon

Associate Professor of Sociology and Public
Health
Editorial Board, *Society and Mental Health*
University of Texas at Dallas
soyoung.kwon@utdallas.edu

Richard Scotch

Professor Emeritus of Sociology,
Public Health, and Public Policy
University of Texas at Dallas
richard.scotch@utdallas.edu

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